

Remote repositories

remote $\Rightarrow$ ptr to version of proj hosted on server, github, gitlab, Bitbucket or even own server.

$\hookrightarrow$.git/config - in local it stored here.

\$git remote $\rightarrow$ list of remotes.

(-v) $\rightarrow$ detailed with urls.

\$git remote add <name> <url>
 $\downarrow$ $\downarrow$
remote name repo-url
usually origin

modify existing :-

$\hookrightarrow$ \$git remote set-url <name> <new-url>

remove :

$\hookrightarrow$ \$git remote remove <name>

Git Fetch :-

$\hookrightarrow$ retrieve updates from rem. repo without automatic merge into local branch.

$\hookrightarrow$ how? 1) connect remote

2) Download commits & stores in .git dir

3) update tracking branches

$\hookrightarrow$ usage :

\$git fetch <rem-name>

\$git fetch <rem-name> <br-name> - spec branch.

\$git diff <loc-br-name> <rem>/<br-name> - diff b/w loc. rem.

\$git fetch (-tags) $\rightarrow$ useful for versioning / releases.

Git pull :-

$\hookrightarrow$ 2 operations $\left\{ \begin{array}{l} \text{git fetch} \\ \text{git merge} \end{array} \right.$

$\hookrightarrow$ how? 1) Identify remote (by url)

2) fetch changes

3) merge changes

checkout feature
\$ git pull --rebase origin main

↳ from feature branch, & keep history clean.

configure pull strategies:-

1) \$ git config --global pull.rebase true

↳ rebase/merge by def.

2) \$ git branch --set-upstream-to=origin/main

↳ when frequently pull - set upstream to simplify

3) \$ git pull --ff-only

↳ only pull if merge fastforwardly.

Git push:-

↳ upload loc. content to remote.

\$ git push <remote> <branch>

↓
usually origin

↓
br want to push.

↳ How?

1) authentication to give perm.

2) check diff - only new cnts / obj transfered

3) update reference -

use:

\$ git push -u origin <upstream-br>

↳ for setting 1st time then
git push enough

→ to see curr tracking branches → \$ git branch -vv

→ conflicts? fetch, merge/rebase, push.

→ commands:

--force → overwrite cnts

--force-with-lease → safe, push only if rem not updated.

--dry-run → see what happen when pushed, w/out actual push.

--tags → push loc tags to rem repos

--set-upstream → set ups. tracking rel. b/w loc & rem.

--delete → del branch on rem rep.